



Indian Association for the Cultivation of Science
(Deemed to be University under *de novo* Category)
Integrated Bachelor's-Master's Program
End-Semester Examination-Autumn 2025

Subject: Chemical Reactivity
Full Marks: 50

Subject Code(s): CHS 2101
Time Allotted: 3 h

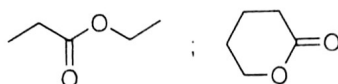
Section-A

(There are FIVE questions in this Section. Answer Q1 and any THREE from Q2-Q5.)

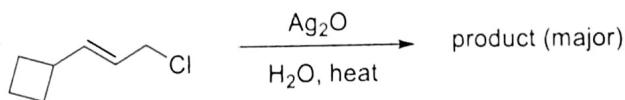
1. Answer ANY FIVE from the following questions.

[2 x 5 = 10]

(a) Which compound among the given options is more susceptible to nucleophilic addition reactions? Justify your answer.

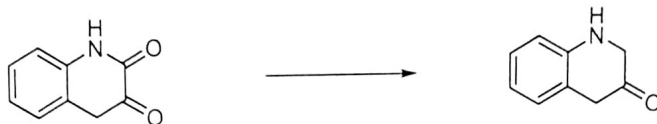


(b) Predict the major product of the transformation given below and provide a suitable reaction mechanism.



(c) Propyl bromide reacts with NaOH in DMSO faster than in water, while *tert*-butyl bromide shows the opposite behavior.

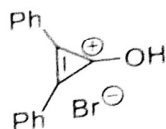
(d) Suggest the most suitable reagent(s) along with the correct reaction sequence to perform the given transformation.



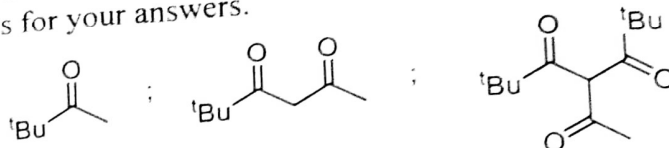
(e) The reaction of ($\pm$)-glycerol with benzaldehyde produces a 6-membered cyclic acetal that includes a mixture of stereoisomers. Draw all possible stereoisomers.

(f) Can *ortho*, *meta*, and *para*-chlorotoluene be distinguished by their ^1H NMR spectra? Justify your answer.

(g) Despite the strain from internal bond angles of only 60° , the given salts can be easily prepared and exhibit exceptional stability. Explain



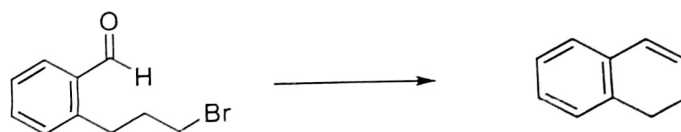
2. (a) Arrange the following ketones in order of increasing enol content in hexane. Provide justifications for your answers. [1.5]



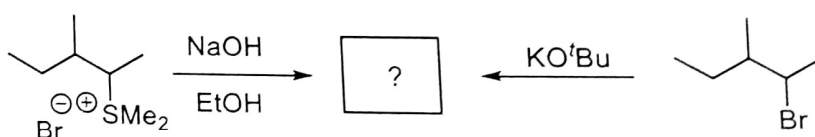
(b) Describe anti-aromaticity with a proper example. [1.5]

(c) The reaction of erythro-3-bromo-2-butanol with hydrobromic acid produces *meso*-2,3-dibromobutane as the major product. Justify this observation with a proper reaction mechanism. [2]

3. (a) Propose the shortest synthetic route for the given conversion by predicting appropriate reagents in the correct reaction order. [2.5]

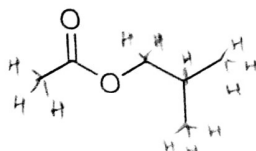


(b) The major product of the reactions given below is identical. Draw the correct structure of the product and explain the observation with a proper reaction mechanism and justification. [2.5]

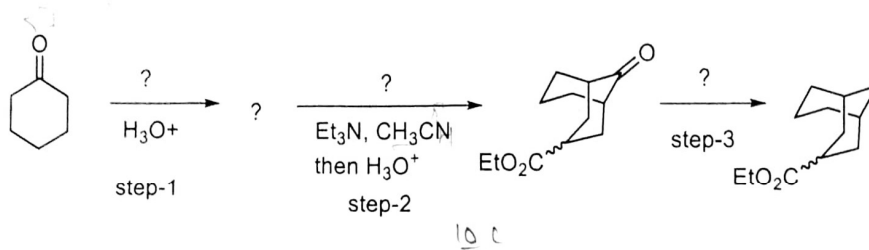


4. (a) Compound **A** ($C_8H_6ClNO_3$) reacts with hydroxylamine in ethanol to produce a mixture of diastereomeric product **B** ($C_8H_7ClN_2O_3$). When product **B** is treated with a strong base, it cyclizes to form product **C** ($C_8H_6N_2O_3$) in moderate yield. Predict the structures of compounds **A-C**, and justify your answer with appropriate reaction mechanisms. [3]

(b) Predict the 1H NMR spectrum of the ester given below. [2]



5. Complete the following transformation by predicting suitable structures for the product(s), reactant(s), and reagent(s). Provide the reaction mechanism for steps 2 and 3. [2+3]



Section-B

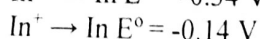
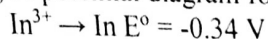
(There are FIVE questions in this Section. Answer Q6 and any THREE from Q7-Q10.)

6. (a) What is the electron pair and molecular geometry of the BrF_5 molecule? Mention different symmetry elements present in the molecule. What is the point group of this molecule? [2]

(b) Write down the formula of diaquabis(ethylenediamine)chromium(III) nitrate. How many isomers are possible for the coordination entity? Draw the structures of each isomer. [2]

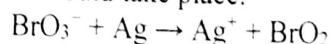
(c) Write down the IUPAC name of $\text{K}[\text{fac-RuCl}_3(\text{PPh}_3)_3]$. Draw the structure of the complex. [1.5]

(d) A potential diagram for indium at $\text{pH} = 0$ shows the following E° values: [1.5]



What is the value of E° for the reduction of In^{3+} to In^+ ?

(e) Balance the following redox reaction and suggest under what condition (acidic or basic) the reaction would take place: [1.5]



① A metal ion in an octahedral field contains five unpaired electrons. If the mean P-value is 20500 cm^{-1} and the $10Dq$ is 15500 cm^{-1} , what will be the spin state of the metal complex? Explain. [1.5]

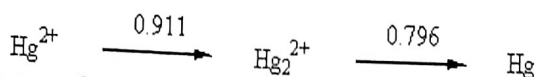
7. (a) Draw the Lewis structure of SCN^- . This ion generally prefers to bind metals in a nonlinear manner, but in some cases forms linear M-SCN bonds. Explain the two binding modes of the ion in terms of their Lewis structure. [2.5]

(b) Write down the Lewis dot formula of the oxo acid, iodic acid (HIO_3). What is the oxidation state of iodine in the oxo acid? Consider the cation of iodine in that oxidation state and show the formation of the oxo acid. Does the bonding between I and O match that obtained from the Lewis structure of the conjugate base of HIO_3 ? [2.5]

8. (a) Draw the d -orbital splitting diagram of $\text{K}_2[\text{CoCl}_4]$ and find out the number of unpaired electrons present in the complex. [2]

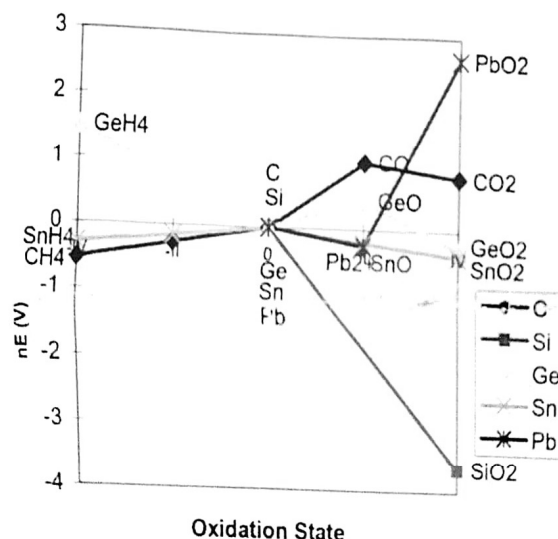
(b) A complex $\text{Co(en)}_2(\text{NO}_2)_2\text{Cl}$ (en = ethylenediamine) has been prepared in several isomeric forms. One form undergoes no reaction with AgNO_3 or en and is optically inactive. A second form reacts with AgNO_3 but not with en and is optically inactive. The third form is optically active and reacts with both AgNO_3 and en. Identify each of these isomeric forms. Name (with the absolute configuration of optical isomers) and sketch each structure. [3]

9. (a) Draw a Frost diagram for mercury using the following Latimer diagram in an acidic solution:



Comment on the tendency of any of the species to act as an oxidizing agent, a reducing agent, or to undergo disproportionation. [2.5]

(c) Consider the Frost diagram for the Group 14 elements in an acid solution and answer the following questions. [2.5]



- Which species among all Group 14 species is the strongest reducing agent?
- Which species among all Group 14 species is the strongest oxidizing agent?
- What reaction is expected to occur when elemental tin is reacted with aqueous acid?
- What should happen when carbon monoxide is bubbled into an acidic solution?
- Write a balanced redox equation for the reaction of elemental silicon and lead dioxide.

10. (a) Calculate the reduction potential of perbromate to bromate at pH 8 with each in 1M concentration. ($E^0_{1/2}(\text{BrO}_4^-/\text{BrO}_3^-) = 1.85 \text{ V}$ at pH 0) [1.5]

(b) Draw the MO diagram of NeO and comment on its bond order. [2]

(c) The pH of 0.1 M $\text{Al}(\text{NO}_3)_3$ in aqueous solution is 3.07. Explain. [1.5]

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